MZUZU UNIVERSITY

FACULTY OF SCIENCE TECHNOLOGY AND INNOVATION

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

BACHELOR OF SCIENCE IN INFORMATION AND COMMUNICATION TECHNOLOGY

ICT 1102: INTRODUCTION TO PROGRAMMING

SUPPLEMENTARY EXAMINATION

DATE: 30th April, 2018 **TIME ALLOWED: 3 HOURS**

INSTRUCTIONS

1. This paper contains **FIVE** questions.
2. Answer **all** questions.
2. Marks for each question are indicated.
3. Write your answers clearly.
4. Indicate the question number for each answer.
5. This paper contains **TWO** pages. Please check.

Question One

a) Explain and give ***one*** example of each of the following language models:

i) Imperative languages. **[3 marks]**

ii) Functional languages. **[3 marks]**

b) Answer the following questions about structured programming.

i) Explain *Structured Programming* as a programming methodology. **[3 marks]**

ii) What are the ***three*** *main advantages* of structured programming over unstructured programming? **[3 marks]**

c) Answer the following questions concerning *language generations*.

i) Name ***two*** *language generations* that allowed users to write programs that were *machine dependent*. **[2 marks]**

ii) State ***two*** differences between the given language generations in ***c(i)*** above. **[2 marks]**

d) Mention any ***five*** key questions to ask when evaluating programming languages for use in a project. **[5 marks]**

Question Two

a) Mention ***four*** reasons why the use of functions and procedures assist in program design. **[4 marks]**

b) Distinguish between the following terms:

i) Procedure and function. **[2 marks]**

ii) Global variable and local variable. **[2 marks]**

iii) Actual parameter and formal parameter. **[2 marks]**

c) Explain the stages of the software testing life cycle. **[14 marks]**

Question Three

a) Briefly explain how linked lists work and differentiate any **two** categories of linked lists? **[6 marks]**

b) Write a data structure that depicts the data as follows: Information on customer orders is stored up in an order file. Each order consists of a header record holding the order number and the customer number, followed by a group of detail records. Each detail record carries the amount debited for a different product. We are required to produce a report showing the total amount debited for each order. **[7 marks]**

c) Illustrate with the use of a pseudo code how global and local variables are implemented in a program. Your illustration should include passing by value. **[10 marks]**

d) Describe any **three** aspects of software that could be considered for testing by the test engineer. **[6 marks]**

Question four

a) Write a pseudo code for the program that accepts five exam grades from the user and computes the average of the grades. Output the average of the five exams. Together with the average, also include a statement "Well done!" in the output if the average is greater than or equal to 60, otherwise output "You need to pull up your socks!" **[6 marks]**

b) Write a pseudo code program to find the *arithmetic mean* of a list of *positive* numbers, and print the arithmetic mean. The list of numbers is not known in advance. Terminate the input of numbers with *zero*. **[5 marks]**

c) Write a pseudo code for the program that calculates the difference of two numbers if the first is greater than the second, otherwise calculates the sum. **[5 marks]**

d) Present the solution in Question four **c)** above in form of a structure diagram. **[5 marks]**

Question Five

a) Write a pseudo code for the program that prompts the user to enter a starting number between 1 and 100 and displays the entered number to the screen then calculates the sum of the numbers 10 to 200 and displays the result to the screen. **[10 marks]**

b) Write a data structure diagram for the following pseudo code extract:

```
Begin A
  if <condition 1> then
        do B;
  else if <condition 2> then
        do C;
        do D;
  endif
```

[3 marks]

c) Differentiate between a stack and a queue as regards the delete operation to remove an element in each. **[2 marks]**

d) Create a program that prints your name a hundred times. Do three versions of this program using a while loop, a do-while loop and a for-loop. **[9 marks]**

END OF QUESTION PAPER